London Ambulance Service NHS
NHS Trust

Support the management of trauma in the Emergency and Urgent Care Setting

Component 8

Musculoskeletal Injuries

Musculoskeletal Injuries – Component 8

OBJECTIVE

- Understand signs and symptoms, complications and treatment for Musculoskeletal injuries.
- Explain interventions that may be required
- Describe time critical injuries and when action should be taken.

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Injuries involving the musculoskeletal (M/S) system are very common and include injuries from sprains to fractures.

Patients with injuries to the musculoskeletal system are likely to be distressed and in pain, therefore consider, when clinically possible, early pain management.



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Mechanism of injury (MOI)



It is Important that you determine the exact mechanism of injury.

A good way to do this is to produce a mental image of the **DIRECTION**, **MAGNITUDE** and **DURATION OF FORCE** that was applied to the injured limb or joint.

Evidence says that a good review of the MOI can help to predict injury patterns.

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Direct force

A bone breaks at the point of impact when the force is too great for the soft tissues to absorb and dissipate it

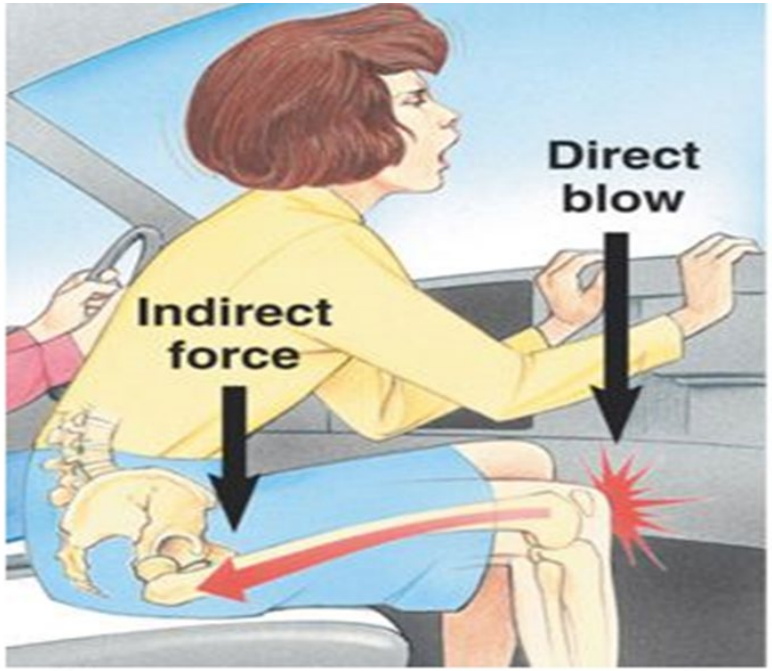


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Indirect force

A force applied to a region of the body is transmitted through the skeleton until it reaches the structurally weakest point causing a bone or bones to #



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REMEMBER

Do **NOT** be distracted from assessing less visible but life threatening problems, however dramatic the M/S injuries are in appearance.



For example:

- **Airway obstruction**
- **Compromised breathing**
- **Spinal injury**

Remember your

C A c B C's

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Musculoskeletal injuries

- Amputations / Partial Amputations
- Fractures (#)
- Dislocations
- Strains
- Sprains



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Amputations, partial amputations and de-gloving

Re-implantation following amputation or reconstruction following partial amputation MAY be possible, as long as the amputated parts are maintained and transported in the best condition possible :

The Amputated Body Part

- Remove any gross contamination
- Cover the part(s) with a moist field dressing
- Secure in a sealed plastic bag
- Place the bag on ice – DO NOT place body parts in direct contact with the ice as this can cause tissue damage, the aim is to keep the temperature low and not freezing

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Amputations, partial amputations and de-gloving

The Patient

- DO NOT irrigate grossly contaminated wounds with saline
- Immobilise a partially amputated limb in a position of normal anatomical alignment
- Where possible dress the injured limb to prevent further contamination
- Apply a saline soaked dressing covered with a occlusive layer
- Pain management, if required – Entonox or ask a senior clinician for advanced pain management.

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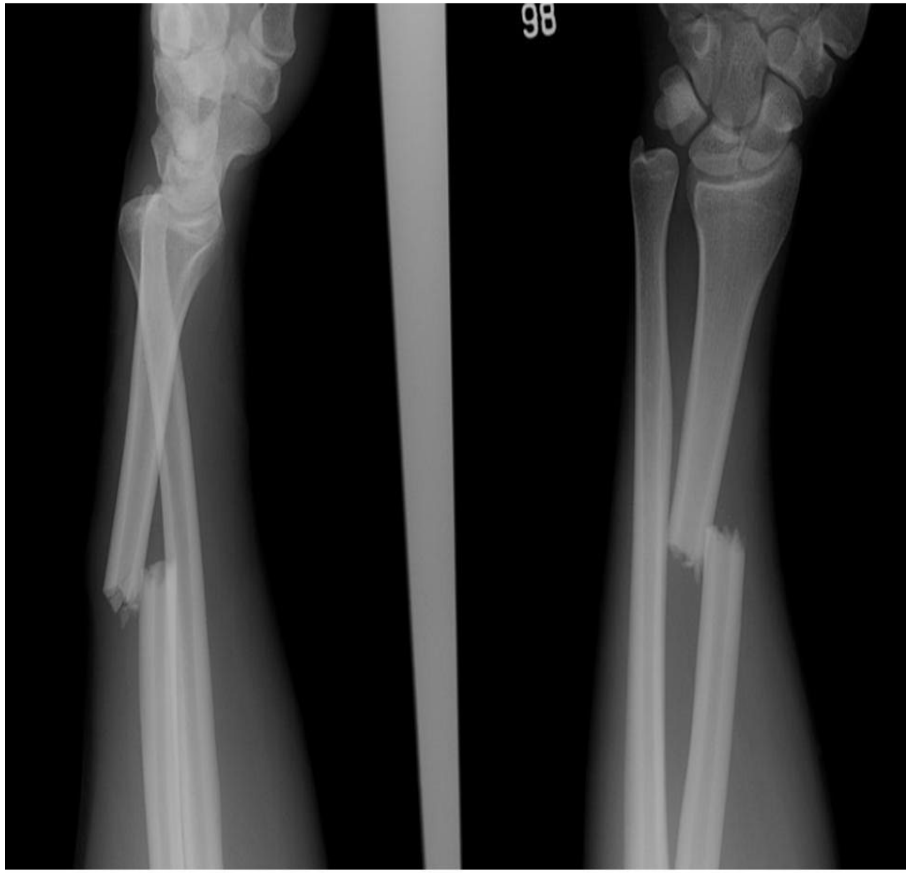
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Fracture :

A break or breaks in the bone…

or where

The level of kinetic energy is great enough to create a deformation of a bone or bones.



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Signs and symptoms

- Pain
- Swelling
- Erythema (redness)
- Deformity



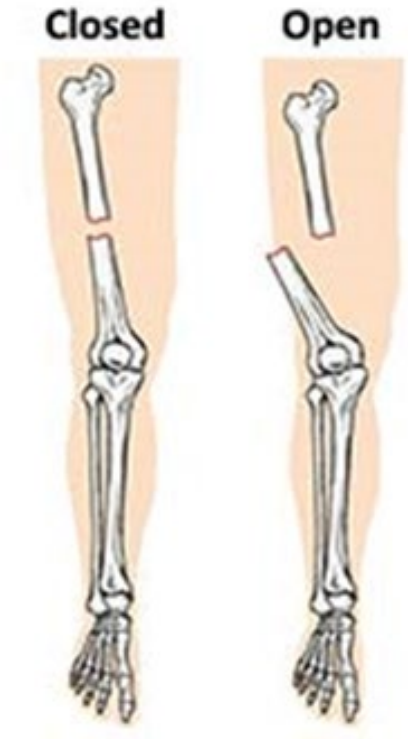
Are all associated with injuries to the M/S system and can usually be easily identifiable.

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Types of Fractures

There are 2 main types of fracture.
These are open and closed.



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Closed

Sometimes called a simple fracture.

A fracture where no puncturing of the skin has occurred.



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Compound # (Open)

A fracture with an wound at the site of injury, caused by bone pushing through the skin.



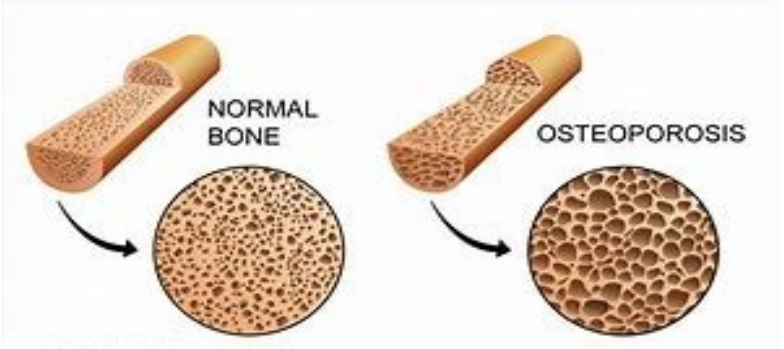
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Pathological Fractures

In older people injuries can occur from relatively minor trauma (e.g. falls from standing height can lead to femoral fracture.)

Accompanying illnesses such as cancer that involve bones (breast , lung and prostate) or osteoporosis, may result in fractures from minor injuries.



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Dislocations

The abnormal separation of joint surfaces.

May occur in isolation or be associated with a fracture.

The muscles involved then spasm causing the joint to lock.





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Sprains and Strains

Defined as damaged to the “soft tissues, ligaments and tendons around a joint without penetrating the joint capsule.”

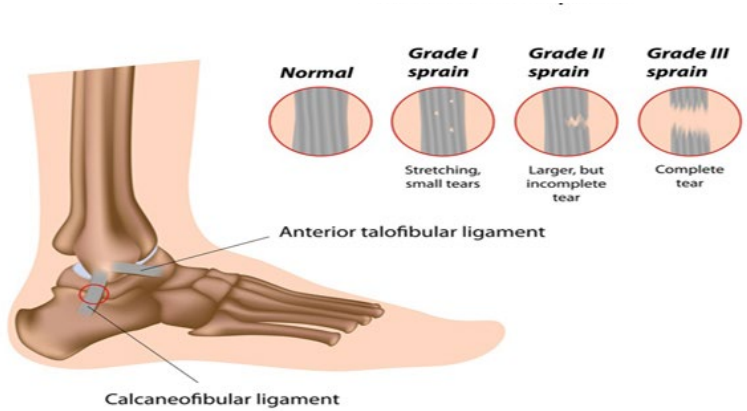


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Sprain

Sprains are ligamental injuries that involve a partial or complete tear of a ligament.



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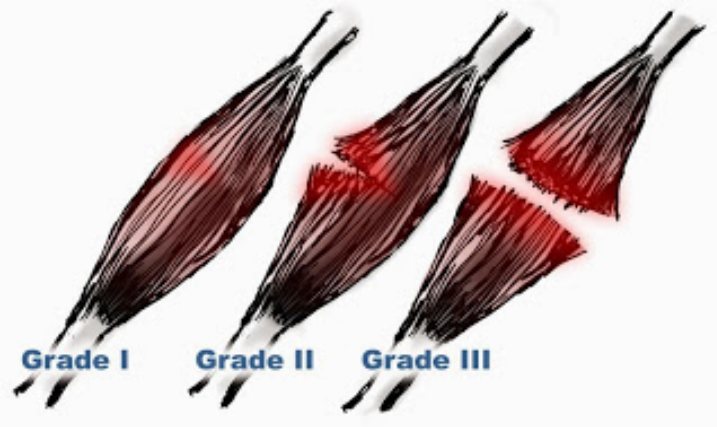
A sprain in the pre hospital setting is difficult to differentiate from a fracture, therefore **ASSUME** there is a fracture and immobilise accordingly.



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Strain

A strain (or 'pull') is a stretch and/or tear of muscle fibers and/or tendon.



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Management

- Assess <C> ABCDE
- Are there any TIME CRITICAL FEATURES
- Correct C ABC
- If **TIME CRITICAL** refer to MAJOR TRUAMA TOOL and transfer to major trauma centre (continue patient management on route)
- Ascertain MOI
- Assess limbs for Motor Sensory Circulation MSC
- Remove jewellery or constrictive clothing
- Consider pain management
- Examination of Injury – Look/Feel/Move
- Immobilise fractures/Apply splintage
- For strains and sprains consider a cold compress and limb elevation.



JRCALC PLUS – Guidelines – Limb Trauma

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Basic principles of **limb immobilisation**:

- Assessment and reassessment of the neurovascular status before and after any manipulation or handling of the fracture using MSC
  - Motor: Ask the patient to move the limb.
  - Sensory: Apply light touch to evaluate sensation.
  - Circulation: Assess a distal pulse and skin temperature
  - Immobilisation of the joints above and below the fracture.



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The six ‘P’s of ischaemia



Sign	Symptom
Pain	Out of proportion to the injury not eased by splinting or pain relief
Pallor	Compromised blood flow to limb
Paralysis	Loss of movement
Paraesthesia	Changes in sensation
Pulselessness	Loss of peripheral pulses which can lead to complete occlusion of circulation
Perishing cold	Limb is cold to touch

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Splinting Reduces:



- Pain
- Haemorrhage and damage to blood vessels and nerves
- Further damage from bone fragments
- Provides support
- Risk of fat embolism
- Pressure on skin and adjacent neurovascular structures
- Muscle spasms





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Injury	Splintage type
# Neck of Femur(NOF)	Padding between legs. Figure of eight around ankles. Broad bandage 2 above 2 below knee
# Shaft of Femur	Kendrick Traction splint. Not be used if # to ankle,tibia,fibula, knee on the same side as femoral #,NOF
# or # Dislocation around the knee	Long Leg Box splint Vacuum splint
Patella Dislocation	Vacuum splint Long leg Box splint. Sometimes when splinting the leg the patella spontaneously relocates



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Injury	Splintage type
Tibia/Fibula shaft #	Long leg box splint Long leg vacuum splint
Ankle #	Short leg box splint Short leg vacuum splint
Foot #'s	Short box splint Short vacuum splint
Clavicle Humerus Radius Ulna	Self splintage Triangular sling Vacuum splint for forearm # Short box splint



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The traction splint should be applied quickly if it does not delay transfer, otherwise apply manual traction, once applied this should not be released.



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Mid shaft femoral fractures (#) are **Time Critical** as blood loss can be considerable (500-2000mls) more if an open #

1000mls plus can be lost from a pelvic #



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Complications of musculoskeletal injuries

- Bleeding
- Damage to blood vessels , nerves , muscles and arteries
- Infection
- Loss of tissue
- Long term disability
- Compartment syndrome





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Summary

- Patients with musculoskeletal injuries/limb trauma may be in considerable pain and distress
- Splinting will contribute to “circulation” care by reducing further blood loss and pain, while on route to hospital
- Do not be distracted from assessing less visible but life threatening problems when dealing with a Musculoskeletal injuries
- Consider the need to request the appropriate clinician with regard to TIME CRITICAL Injuries and PAIN management.

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Any Questions?



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